

Planckd2025 Quantum Algorithm Track — A Complete Analytical and Simulation Framework (Problems 0–8)

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Abstract

This report documents the theoretical foundations, the algorithmic approaches and the full mathematical derivations behind Problems 0–8 carried out in the Planckd2025 Quantum Algorithm Track. Each problem is presented with: a concise theoretical background, a clear description of the approach used for simulation/implementation, and step-by-step mathematical derivations that justify the approaches and explain the observed behaviors. The goal is a self-contained technical record so that a reader with basic quantum computing background can follow the reasoning and reproduce the analytical steps without referring to code.

Notation

We use Dirac notation for states ($|\cdot\rangle$), standard Pauli matrices X, Y, Z , and identity I . Expectation values are written $\langle\cdot\rangle$. Probability amplitudes are complex. Subscripts like $_c$ indicate coin qubit; $_p$ position register when needed.

Problem 0 — Classical Random Walk (Theory, Approach, Mathematics)

Theory

A one-dimensional unbiased classical random walk (CRW) is a stochastic process where at each discrete time step the walker moves either left or right with probability $1/2$. After t independent steps the walker position is the sum of independent ± 1 random variables. CRW provides the reference diffusive scaling against which quantum walks are compared.

Approach

We consider ensemble averages (many independent walkers) and compute the probability distribution $P(x, t)$ and the second moment $\langle x^2 \rangle$. Numerically one simulates many sample walks and averages; analytically use binomial-to-normal approximation for large t .

Mathematical derivation

Let $X_k \in \{-1, +1\}$ be IID random variables with $\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = 1/2$. Position after t steps:

$$X(t) = \sum_{k=1}^t X_k. \quad (1)$$

Mean and variance:

$$\mathbb{E}[X(t)] = \sum_{k=1}^t \mathbb{E}[X_k] = 0, \quad (2)$$

$$\text{Var}[X(t)] = \sum_{k=1}^t \text{Var}[X_k] = t \cdot 1 = t. \quad (3)$$

The Central Limit Theorem yields for large t :

$$P(x, t) \approx \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right). \quad (4)$$

The Root-Mean-Squared (RMS) displacement:

$$\sigma_{\text{CRW}}(t) = \sqrt{\langle X(t)^2 \rangle} = \sqrt{t}. \quad (5)$$

Hence classical spreading is diffusive, scaling as $\mathcal{O}(\sqrt{t})$.

Problem 1 — Quantum Coin Flip (Theory, Approach, Mathematics)

Theory

A quantum coin flip using the Pauli- X operator produces deterministic flip dynamics on the coin qubit. If the coin is used to control a shift on a position register, but the coin itself is never placed into a superposition, the walk remains effectively deterministic.

Approach

We model two registers: coin qubit $\mathcal{H}_c$ and a small position register $\mathcal{H}_p$. Apply X on the coin and then apply controlled-NOT-like shifts that move the walker conditioned on the coin. Because X maps $|0\rangle \leftrightarrow |1\rangle$ deterministically, the direction alternates each step.

Mathematical derivation

One step operator:

$$U_{\text{step}} = \text{Shift}_{c \rightarrow p} \cdot (X_c \otimes I_p). \quad (6)$$

If initial state is $|x=0\rangle \otimes |0\rangle_c$, after one step coin becomes $|1\rangle$ and position shifts right; next step coin flips back to $|0\rangle$ and position shifts left. There is no interference term since the coin state never forms a coherent superposition between the alternatives. Probabilities remain 0 or 1 along deterministic path.

Problem 2 — The Superposed Walker (Theory, Approach, Mathematics)

Theory

Placing the coin qubit in superposition introduces quantum interference between different paths. For the discrete-time coined quantum walk (DTQW) on a line, repeated coin and shift operations create an amplitude profile that spreads ballistically due to interference.

Approach

Use a single coin qubit with Hadamard coin H and a conditional shift S acting on the position register. At each time step apply $U = S \cdot (H \otimes I)$. Start in $|x = 0\rangle \otimes |0\rangle_c$ (or coin in $|+\rangle$) and evolve t steps. Extract position marginal probabilities by tracing out coin or computing probabilities over position basis, then compute RMS.

Mathematical derivation

State expansion at time t :

$$|\Psi(t)\rangle = \sum_{x \in \mathbb{Z}} (a_x(t) |x\rangle \otimes |0\rangle_c + b_x(t) |x\rangle \otimes |1\rangle_c). \quad (7)$$

Coin operator (Hadamard):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (8)$$

Shift operator S acts as

$$S|x\rangle \otimes |0\rangle_c = |x-1\rangle \otimes |0\rangle_c, \quad S|x\rangle \otimes |1\rangle_c = |x+1\rangle \otimes |1\rangle_c. \quad (9)$$

One-step recursion for amplitudes:

$$a_x(t+1) = \frac{1}{\sqrt{2}} a_{x+1}(t) + \frac{1}{\sqrt{2}} b_{x+1}(t), \quad (10)$$

$$b_x(t+1) = \frac{1}{\sqrt{2}} a_{x-1}(t) - \frac{1}{\sqrt{2}} b_{x-1}(t). \quad (11)$$

These coupled recurrences yield constructive interference at ballistic fronts. The position probability is

$$P(x, t) = |a_x(t)|^2 + |b_x(t)|^2. \quad (12)$$

Compute second moment:

$$\sigma_{\text{QW}}^2(t) = \sum_x x^2 P(x, t), \quad (13)$$

and one finds asymptotically (see standard DTQW analysis) that

$$\sigma_{\text{QW}}(t) \sim ct \quad (t \rightarrow \infty), \quad (14)$$

for constant c depending upon coin; for Hadamard coin c is nonzero producing ballistic $\mathcal{O}(t)$ growth. Detailed solution can be obtained via Fourier methods: define Fourier transforms

$$\tilde{a}_k(t) = \sum_x a_x(t) e^{-ikx}, \quad \tilde{b}_k(t) = \sum_x b_x(t) e^{-ikx}, \quad (15)$$

derive 2×2 evolution matrix in k -space and diagonalize to extract group velocities producing linear spreading.

Problem 3 — Graph-Based Quantum Walk (Theory, Approach, Mathematics)

Theory

Quantum walks on graphs generalize 1D walks to arbitrary topology. Continuous-time quantum walks (CTQW) use graph Laplacian/adjacency matrix as the Hamiltonian; discrete-time walks use coin and shift generalized to graph edges. CTQW and DTQW both exploit interference to produce different transport properties than classical random walks on graphs.

Approach

We consider CTQW on the cycle graph C_N as a representative example and compare hitting probabilities to classical Markov chain results. Use adjacency matrix A and Hamiltonian $H = -\gamma A$ with rate γ .

Mathematical derivation

Let A be adjacency of C_N so that $A_{i,i\pm1} = 1$. CTQW evolution:

$$|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle, \quad H = -\gamma A. \quad (16)$$

Spectral decomposition: $A = \sum_m \lambda_m |v_m\rangle\langle v_m|$, then

$$|\psi(t)\rangle = \sum_m e^{i\gamma\lambda_m t} \langle v_m | \psi(0) \rangle |v_m\rangle. \quad (17)$$

On cycles eigenvectors are plane waves; amplitudes interfere periodically causing peaks in hitting probabilities that can exceed classical stationary probabilities at certain times. Classical Markov chain mixing time scales differently, often slower for certain graphs.

Problem 4 — Quantum State Tomography (Theory, Approach, Mathematics)

Theory

Quantum State Tomography aims to reconstruct an unknown density matrix ρ by performing informationally complete measurements. For two qubits, measuring in the tensor-product Pauli basis provides a full linear system.

Approach

We measure expectation values of $\sigma_i \otimes \sigma_j$ for $i, j \in \{I, X, Y, Z\}$, estimate coefficients, reconstruct ρ via linear inversion, and when necessary project onto the nearest physical (positive semidefinite, unit-trace) state using maximum-likelihood or principal-eigenvector projection for pure-state estimation.

Mathematical derivation

Express ρ as:

$$\rho = \frac{1}{4} \sum_{i,j \in \{I,X,Y,Z\}} c_{ij} (\sigma_i \otimes \sigma_j), \quad (18)$$

with coefficients

$$c_{ij} = \text{Tr}(\rho(\sigma_i \otimes \sigma_j)). \quad (19)$$

Given estimated expectation values $\hat{c}_{ij}$, construct

$$\hat{\rho} = \frac{1}{4} \sum_{i,j} \hat{c}_{ij} (\sigma_i \otimes \sigma_j). \quad (20)$$

To obtain a physical pure-state estimate, compute eigen-decomposition $\hat{\rho} = V\Lambda V^\dagger$ and take principal eigenvector $v_{\max}$ corresponding to largest eigenvalue; estimated state $|\hat{\psi}\rangle = v_{\max}$.

Problem 5 — Quantum Oscillator Walk (Theory, Approach, Mathematics)

Theory

Mapping position indices to quantized energy levels yields dynamics analogous to quantum harmonic oscillator transitions. By engineering coin and shift operations with amplitudes resembling creation/annihilation couplings, walker dynamics explore energy-level space.

Approach

We model a two-level 'internal' coin per energy level mapping to basis $|n\rangle \otimes |q\rangle$ with $q \in \{0, 1\}$. The unitary step couples neighboring n levels with amplitudes reflecting ladder-operator-like coefficients; phase potentials $\phi(n)$ encode external fields.

Mathematical derivation

Define step Hamiltonian (heuristic):

$$H_{\text{step}} = \sum_n (g_n |n+1\rangle \langle n| + g_n^* |n\rangle \langle n+1|), \quad (21)$$

with coupling amplitudes g_n (choose $g_n = \sqrt{n+1}$ to mimic oscillator matrix elements if desired). Adding phase potential:

$$U_V = \exp \left(i \sum_n \phi(n) |n\rangle \langle n| \right), \quad \phi(n) = \alpha n \text{ or } \alpha n^2. \quad (22)$$

Time evolution per step:

$$|\psi_{t+1}\rangle = U_{\text{step}} U_V |\psi_t\rangle. \quad (23)$$

Compute site probabilities $P(n, t) = \langle n | \text{Tr}_q(|\psi_t\rangle \langle \psi_t|) | n \rangle$ and the RMS over n :

$$\sigma_n^2(t) = \sum_n n^2 P(n, t), \quad (24)$$

whose growth characterizes ballistic vs localized behavior depending on $\phi(n)$.

Problem 6 — Adiabatic Quantum Computing (AQC) (Theory, Approach, Mathematics)

Theory

AQC prepares the ground state of a problem Hamiltonian H_P by slowly interpolating from an easy initial Hamiltonian H_B . The adiabatic theorem guarantees remaining in the ground state provided the evolution is slow relative to the inverse square of the minimum spectral gap.

Approach

We choose a linear schedule $s(t) = t/T$ and study instantaneous eigenvalues $E_k(s)$ of $H(s)$ to locate $\Delta_{\min} = \min_s (E_1(s) - E_0(s))$. Estimate runtime T necessary to achieve high ground-state fidelity. Numerical diagonalization for small instances confirms theoretical scaling.

Mathematical derivation

Interpolating Hamiltonian:

$$H(s) = (1-s)H_B + sH_P, \quad s \in [0, 1]. \quad (25)$$

Adiabatic condition (heuristic bound):

$$T \gg \max_{s \in [0,1]} \frac{|\langle E_1(s) | \partial_s H(s) | E_0(s) \rangle|}{\Delta(s)^2}, \quad (26)$$

where $\Delta(s) = E_1(s) - E_0(s)$ is the instantaneous gap. For many practical cases, a conservative rule-of-thumb:

$$T \propto \frac{1}{\Delta_{\min}^2}. \quad (27)$$

Therefore exponentially small gaps imply exponential runtime; polynomial gaps yield polynomial runtime scaling.

Problem 7 — QAOA (Theory, Approach, Mathematics)

Theory

QAOA is a variational algorithm that approximates adiabatic evolution via p alternating layers of unitary operators generated by H_P and mixing Hamiltonian H_B . Classical optimizers tune (γ, β) to maximize the objective expectation.

Approach

Construct the parameterized circuit:

$$|\psi_p(\gamma, \beta)\rangle = \prod_{k=1}^p e^{-i\beta_k H_B} e^{-i\gamma_k H_P} |+\rangle^{\otimes n}. \quad (28)$$

Evaluate cost

$$C(\gamma, \beta) = \langle \psi_p | H_P | \psi_p \rangle \quad (29)$$

and use classical optimization to find parameters maximizing C .

Mathematical derivation

For small instances (e.g. MaxCut on 3 nodes), H_P can be written explicitly and expectation values computed analytically at low p by expanding the unitaries or using operator identities. The connection to AQC arises by interpreting γ_k, β_k as discrete time-steps of a Trotterized adiabatic evolution:

$$\gamma_k \approx s_k \Delta t, \quad \beta_k \approx (1 - s_k) \Delta t, \quad (30)$$

where $\Delta t = T/p$ and s_k is schedule value at step k . For many problems QAOA at small p attains high-quality approximations due to non-linear parameter landscapes that exploit problem symmetries.

Problem 8 — Open Quantum Systems & Noise Robustness (Theory, Approach, Mathematics)

Theory

Quantum computations on NISQ hardware are affected by noise channels such as amplitude damping and dephasing. Modeling these effects via Kraus operators or Lindblad master equations quantifies their impact on algorithmic performance.

Approach

We model noise via Kraus maps applied after unitary layers (stroboscopic noise model). We analyze how shallow circuits mitigate accumulation of error and how measurement-based adaptive corrections (classical feedback) can harness the Quantum Zeno Effect (QZE).

Mathematical derivation

Amplitude damping Kraus operators:

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1 - p_{AD}} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{p_{AD}} \\ 0 & 0 \end{pmatrix}. \quad (31)$$

Channel action on density matrix:

$$\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger. \quad (32)$$

Dephasing channel for a single qubit:

$$\mathcal{D}_{p_\phi}(\rho) = (1 - p_\phi)\rho + p_\phi Z\rho Z. \quad (33)$$

Adaptive measurement + correction effect: model measurement on a qubit with projectors P_0, P_1 and conditional corrective unitary U_c applied when outcome indicates error. Frequent measurements with immediate correction realize a Zeno-like suppression of transitions. The survival probability in the subspace under repeated projective checks separated by time τ obeys:

$$P_{\text{surv}}(t = n\tau) \approx \left(1 - \frac{\tau^2}{\tau_Z^2}\right)^n \approx \exp\left(-\frac{t\tau}{\tau_Z^2}\right) \approx 1 - \frac{t^2}{\tau_Z^2} \quad (34)$$

for small τ , demonstrating freezing of evolution as $\tau \rightarrow 0$.

Discussion and Conclusion

This document assembled the theoretical background, the simulation/implementation reasoning, and detailed mathematical derivations for Problems 0–8 in the Planckd2025 track. The central thread is the contrast between classical diffusion ($\sigma \propto \sqrt{t}$) and quantum ballistic spreading ($\sigma \propto t$) driven by interference. The report also connects search and optimization (Grover/QAA/QAOA) to walk dynamics and shows how noise models and adaptive corrections influence real-system performance. Together these problems map a pedagogical and research-suitable path from foundations to NISQ-aware algorithmic practice.

— End of Report —